

# Math Camp

## Lecture 6

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# Functions on $\mathbb{R}$

- Economics is inherently a multivariate affair
- Much of the analysis proceeds either by reducing a multivariate problem to a decoupled collection of univariate problems, or by extending univariate results to multivariate settings
- Most of the main ideas and techniques are already present in the univariate world
  - A world that admits graphical approaches

# Continuity

- $f : \mathbb{R} \rightarrow \mathbb{R}$  is *continuous* at  $x \in \mathbb{R}$  provided that whenever  $x_n \rightarrow x$  it follows that  $f(x_n) \rightarrow f(x)$ 
  - $\mathbb{R}$  may be replaced by any subset of  $\mathbb{R}$ , usually open or closed
  - The definition makes sense for functions between two metric spaces, and is what continuity at a point means there
- Equivalently,  $f$  is continuous at  $x$  provided that for any  $\varepsilon > 0$  there is a  $\delta > 0$  such that

$$|y - x| < \delta \Rightarrow |f(y) - f(x)| < \varepsilon$$

# Continuity

- Continuity is a *local* notion: it is defined at a point
  - $f$  is continuous *on its domain* provided it is continuous at every point in the domain
- Continuity preserves some notion of nearness: if  $y$  is near  $x$  then  $f(y)$  is near  $f(x)$ 
  - More precisely, you can make  $f(y)$  near  $f(x)$  by choosing  $y$  sufficiently near  $x$
- If  $f$  is continuous at  $x$  then the behavior of  $f$  near  $x$  can't be too crazy

# Continuity

- Continuity is preserved by many common point-wise transformations
- If  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  are continuous, then so are
  - $f + g$
  - $f \cdot g$
  - $f \circ g$

# Intermediate value theorem

## Intermediate Value Theorem

Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. If  $f(a) > 0 > f(b)$  or if  $f(a) < 0 < f(b)$  then there exists  $c \in (a, b)$  such that  $f(c) = 0$ .

- The most important result on continuous functions from  $\mathbb{R}$  to  $\mathbb{R}$
- Its importance concerns solutions to equations
  - It is an *existence theorem* and provides precise conditions under which an equation is guaranteed to have a solution

# Existence theorems

- Existence theorems are wonderfully powerful as they often require only weak assumptions
- There is a trade off
  - They do not guarantee uniqueness as there may be many zeros of  $f$  in  $(a, b)$
  - They are not typically constructive: they provide no method for constructing the object whose existence is established

# Differentiability

- $f : \mathbb{R} \rightarrow \mathbb{R}$  is *differentiable* at  $x$  provided the following limit, denoted  $f'(x)$ , exists

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

- $\mathbb{R}$  may be replaced by an open set
- Like continuity, differentiability is a local notion
  - $f$  is differentiable if it is differentiable at every point in the domain

# Local linearity

- Continuous functions are reasonably well-behaved but differentiable functions are *very* well-behaved
  - Locally, they act just like linear functions
  - This is why differentiability is so important and calculus so useful
- For small values of  $\Delta x$  the following approximation is pretty good

$$f(x + \Delta x) - f(x) \approx f'(x)\Delta x$$

- Letting  $z = x + \Delta x$ , if  $z$  is near  $x$  then

$$f(z) \approx f(x) + f'(x)(z - x)$$

- $f$  looks like the line  $y = mz + b$  with  $m = f'(x)$  and  $b = f(x) - f'(x)x$

# The differential

- Rewrite the approximation as  $\Delta f \approx f'(x)\Delta x$
- A linear function from  $\mathbb{R}$  to  $\mathbb{R}$  is exactly and only multiplication by a real number
  - So  $f'(x)$  itself defines a linear function  $\mathbb{R} \rightarrow \mathbb{R}$
  - It takes small changes in  $x$  to approximations of the corresponding small changes in  $f$
- Viewed as a linear map, the derivative is the *differential*

$$df = f'(x) dx$$

- $dx$  and  $df$  are *infinitesimals*, loosely thought of as infinitely small
- They have no meaning in isolation, only when written in relation to other differentials

# The differential

- A differentiable function is continuous.

# Rules of differentiation

- Differentiability is preserved under common point-wise transformations: if  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  are differentiable, then so are  $f + g$ ,  $f \cdot g$  and  $f \circ g$
- While the definition of differentiability is somewhat daunting, calculating derivatives is often mechanical — hence the name, *the calculus*
  - $h = f + g \Rightarrow h' = f' + g'$
  - $h = f \cdot g \Rightarrow h' = f' \cdot g + g' \cdot f$
  - $h = f \circ g \Rightarrow h' = (f' \circ g) \cdot g'$

# Mean value theorem

## Mean Value Theorem

Let  $U$  be an open set containing the interval  $[a, b]$ , and let  $f : U \rightarrow \mathbb{R}$  be differentiable. Then there exists  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

- The most important result on differentiable functions

# Mean value theorem

- The theorem says the slope of the line connecting the endpoints of  $f$ , the *secant* line, equals the slope of the tangent line at some point inside  $(a, b)$
- More generally, think of  $f$  as the position of something at time  $t \in [a, b]$ 
  - $f'$  is the instantaneous velocity and the RHS is the average velocity over  $a$  to  $b$
  - At some point in  $(a, b)$  the instantaneous velocity must equal the average velocity
- Its principal importance for economics is that it is a special case of Taylor's theorem

# Higher-order derivatives

- If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is differentiable then  $f'$  may be viewed as a function  $\mathbb{R} \rightarrow \mathbb{R}$ , and it may itself be differentiable
  - If it is,  $f''$  is the *second derivative* of  $f$
  - The process may be repeated so long as the derivative exists
- $f^{(k)}$  denotes the  $k$ th derivative of  $f$
- $f$  is  $C^k$  provided its derivative may be computed  $k$  times and, as a function,  $f^{(k)}$  is continuous

# Integration

- Differentiation allows for the *local* analysis of a function; integration speaks to a measure of its *global* behavior
- Given  $F : [a, b] \rightarrow \mathbb{R}$  we can use differential calculus to think about small changes in  $F$  given small changes in  $x \in [a, b]$ ,

$$dF = f(x) dx$$

- Here  $f$  is a derivative of  $F$
- Conversely, suppose that instead of knowing  $F$ , we know how it changes at each point in an interval
  - We can compute the total change by adding up all the little changes

$$F(b) - F(a) = \sum dF = \sum f(x) dx$$

# Partitions

- A *partition* of an interval  $[a, b]$  is

$$\mathcal{P}_n = \{x_0 = a, x_1, \dots, x_n = b\}$$

with  $x_i < x_{i+1}$

- ▶ It partitions  $[a, b]$  into a collection of intervals with endpoints  $x_i, x_{i+1}$
- ▶ These may be open or closed depending on how you construct the partition
- Given  $\mathcal{P}_n$ , we say  $\hat{x} = \{\hat{x}_1, \dots, \hat{x}_n\}$  is *admissible* if  $\hat{x}_i \in [x_{i-1}, x_i]$

# Refinements

- The pair  $(\mathcal{Q}_m, \hat{y})$  is a *refinement* of  $(\mathcal{P}_n, \hat{x})$  provided
  - $m \geq n$
  - as sets,  $\mathcal{P}_n \subset \mathcal{Q}_m$
  - for each  $i \in \{1, \dots, n\}$  there is a  $j \in \{1, \dots, m\}$  so that  $\hat{x}_i = \hat{y}_j$
- Intuitively,  $(\mathcal{Q}_m, \hat{y})$  dissects the interval more finely than  $\mathcal{P}_n$ , while at the same time respecting  $(\mathcal{P}_n, \hat{x})$ 's cutoffs and admissible points

# Riemann sums

- Given  $f$ ,  $\mathcal{P}_n$  and admissible  $\hat{x}$ , the associated *Riemann sum* is

$$\mathcal{R}(f, \mathcal{P}_n, \hat{x}) = \sum_{i=1}^n f(\hat{x}_i) \Delta x_i$$

where  $\Delta x_i = x_i - x_{i-1}$

- Think of it as a formal approximation to the total change

$$\sum f(x) \, dx \approx \sum_{i=1}^n f(\hat{x}_i) \Delta x_i$$

# Riemann sums

- Given  $[a, b]$  and  $f$  continuous over the interval, the Riemann sum approximates the area between  $f$  and 0 by summing the geometric area of  $n$  partitioned rectangles
  - The rectangles have height  $f(\hat{x}_i)$  and width  $\Delta x_i$
- The area of individual rectangles can be negative if there exists  $c \in [a, b]$  such that  $f(c) < 0$ 
  - The Riemann sum is the area of the rectangles with  $f(\hat{x}_i) > 0$  minus the area of those with  $f(\hat{x}_i) < 0$

# The Riemann integral

- To make the approximation precise, take a limit
- Informally, the *Riemann integral* of  $f$  over  $[a, b]$ , written  $\int_a^b f(x) \, dx$ , is the limit, when it exists, of the Riemann sums, where the limit is taken over refinements
- Formally,  $f$  is *Riemann integrable* on  $[a, b]$  provided there is a real number  $\int_a^b f(x) \, dx$  such that for any  $\varepsilon > 0$  there is a pair  $(P_n, \hat{x})$  with the property that whenever  $(Q_m, \hat{y})$  is a refinement of  $(P_n, \hat{x})$

$$\left| \mathcal{R}(f, Q_m, \hat{y}) - \int_a^b f(x) \, dx \right| < \varepsilon$$

- Continuous functions are Riemann integrable

# Fundamental theorem of calculus

## Fundamental Theorem of Calculus

Let  $g : [a, b] \rightarrow \mathbb{R}$  be continuous. Define  $G : [a, b] \rightarrow \mathbb{R}$  by

$$G(x) = \int_a^x g(s) \, ds$$

Then  $G'(x) = g(x)$ .

- If  $F'(x) = f(x)$  and  $f$  is Riemann integrable then
$$F(b) - F(a) = \int_a^b f(x) \, dx$$

# Total change

- Assume  $F' \equiv f$  is continuous on  $[a, b]$  and suppose we can find  $F : [a, b] \rightarrow \mathbb{R}$  such that  $F'(x) = f(x)$ 
  - By the FTC,  $F(x) = \int_a^x f(x) \, dx + \alpha$
  - Observing that  $F(a) = \int_a^a f(x) \, dx + \alpha = \alpha$ , it follows that

$$F(b) - F(a) = \int_a^b f(x) \, dx$$

- Given the little changes, we need not add them up
  - We can find a function whose derivative gives the little changes and evaluate it at the endpoints

# Integrals in economics

- A common application is to solve differential equations
  - You may know how a variable changes over time and want to compute its value at each point in time
- More generally, it is useful to think of an integral as capturing an *aggregate* or an *average* associated to a given function
  - If a model has a collection of consumers indexed by the unit interval then aggregate consumption is the integral of individual consumption levels over the interval

# Exponents and logs

- The derivative of  $\frac{1}{n} \cdot x^n$  is  $x^{n-1}$  whenever  $n \neq 0$ 
  - What about  $n = 0$ ? What function is it whose derivative is  $x^{-1}$ ?
- Using FTC, for  $x \geq 1$  define

$$\log(x) = \int_1^x \frac{1}{t} dt$$

and for  $x \in (0, 1)$  let  $\log(x) \equiv -\log(1/x)$

- Often referred to as the *natural log* and denoted  $\ln(x)$
- Since it is the main logarithm in economics, we refer to it as *the* logarithm,  $\log(x)$
- The logarithm is a limit of Riemann sums.

# The exponential

- By the fundamental theorem log is differentiable; since its derivative is positive, it is invertible
- Let  $\exp : \mathbb{R} \rightarrow (0, \infty)$  be its inverse. Then

$$\begin{aligned}\log(\exp(x)) = x &\Rightarrow \frac{d \log(\exp(x))}{dx} = \frac{dx}{dx} \\ &\Rightarrow \frac{1}{\exp(x)} \frac{d \exp(x)}{dx} = 1 \\ &\Rightarrow \frac{d \exp(x)}{dx} = \exp(x)\end{aligned}$$

# The exponential

- The exponential function, and its inverse the logarithm, are the most important functions in mathematics
  - They are very likely the only functions you know of that are not polynomials
- They have the following dual properties
  - $\log(xy) = \log(x) + \log(y)$  and  $\exp(x) \exp(y) = \exp(x + y)$
  - $\log(x^{-1}) = -\log(x)$  and  $\exp(-x) = \exp(x)^{-1}$
- These properties allow us to define the more general exponential of a positive real number

$$x^\alpha \equiv \exp(\alpha \log(x)) \quad \text{for } x > 0, \alpha \in \mathbb{R}$$

# Logs in economics

- The uses of logs and exponents in economics are myriad
- If gdp  $y$  grows at a constant rate  $r > 0$ , i.e.  $\dot{y} = ry$ , we may solve for  $y$  as a function of  $t$ 
  - More generally, the exponential function can be defined as the solution to this differential equation

- If  $y = f(x)$ , the *elasticity* of  $y$  with respect to  $x$  is

$$\varepsilon_{yx} = \frac{dy}{dx} \frac{x}{y}$$

- Intuitively, the percent change in  $y$  given a percent change in  $x$
- Letting  $\hat{y} = \log y$  and  $\hat{x} = \log x$ , taking log transforms of the data and then performing a regression gives estimated coefficients that are the elasticities